# CHLOÉ TANAKA

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HARVARD UNIVERSITY

Harvard Kennedy School
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## Education

**Harvard University**
Ph.D. Public Policy, Economics Track, Expected Completion May 2027

**University of Cambridge**
M.Phil. Economics, 2016

**University of Alaska Anchorage**
B.A. Economics, Minor Mathematics, *summa cum laude*, 2013

**Georgetown University**
B.A. Economics, Minor Mathematics, First Honors, Transferred December 2010

## Fields

Primary Field: Environmental and Energy Economics
Secondary Fields: Labor Economics, Education, Public Economics

## References

| **Gordon Hanson (Chair)** | **Joseph Aldy** | **Joshua Goodman** |
|---|---|---|
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## Fellowships & Awards

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|---|---|
| Harvard GSAS Dissertation Completion Fellowship | 2026-2027 |
| Harvard University Graduate Prize Fellowship | 2021-2023 |
| Marshall Scholarship Finalist | 2014 |
| UAA Garth N. Jones Undergraduate Award for Excellence in Writing | 2013 |
| UAA Award for Academic Excellence | 2013 |
| UAA Bradford Tuck Senior in Economics Scholarship | 2011-2012 |

## Teaching

*Principles of Economics – Macroeconomics (ECON 10b),* Harvard University — 2024
Teaching Fellow for Professor David Laibson and Professor Jason Furman

*Principles of Economics – Microeconomics (ECON 10a),* Harvard University — 2023
Teaching Fellow for Professor David Laibson and Professor Jason Furman

## Research Experience

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|---|---|
| Doctoral Researcher, HKS, Reimagining the Economy, Gordon Hanson | 2022-2026 |
| Research Professional, UAA, Institute of Social and Economic Research | 2013-2015 |
| Research Assistant, UAA, Experimental Economics Lab | 2011-2013 |

## Job Market Paper

**Powering the Workforce: The Role of Community Colleges in Energy Transformations**


Energy transformations change the skills local employers require, but does training adapt alongside? I study how training institutions and students respond to energy-induced shifts in local labor demand, using institutional data on the programs and graduates of U.S. community colleges from 1990 to 2024. I focus on two energy transformations, the shale oil and gas boom and the expansion of solar power. In both, resource endowments determine where the transformation occurs, while advances in technology and policy determine when. Shale-exposed colleges expand energy programs by 30%, relative to baseline, and graduates by 117%, primarily through new programs. Taken at face value, this response produced roughly 5,500 additional energy graduates in the subsequent decade, over one third of the estimated boom-induced increase in local middle-skill energy employment. Relative to unexposed colleges, non-energy fields have 10% fewer programs and 31% fewer graduates. In contrast, the solar effect is smaller, even per new job created. A one-gigawatt increase in local solar capacity raises solar programs by 10% and graduates by 21%. Still, colleges adjust on a more granular margin, nearly doubling solar sections in California course-level data. In individual-level records from Texas, energy graduates earn about 14% more over ten years than similar graduates of other technical programs. These findings demonstrate how energy transformations shape local human capital formation through training institutions.

**Publications**

**"Short-run impacts of a severance tax change: Evidence from Alaska."**
*Energy Policy*, Volume 107, 2017 (with Matthew Reimer and Mouhcine Guettabi)

**Working Papers**

**Blowin' in the Wind: The Impact of Climate Disasters on College Majors and Innovation,** *with Avinash Moorthy*

Could extreme weather influence education and career choices? This project investigates how foundational experiences with extreme weather shape human capital formation and innovative activity. We study the causal impact of exposure to extreme weather on college major choices, and on the direction, quality, and intensity of inventors' output, examining whether shifts in educational pathways serve as a mediating channel. We use NOAA records on natural disaster damages to determine the timing and location of exposure. We combine these records with administrative data on educational institutions, college student surveys, LinkedIn employment and education histories, and individual-level patent records. Comparing students and graduation cohorts exposed to major natural disasters with their unexposed counterparts, we find preliminary evidence that exposure shifts major choices toward green fields, with stronger effects for earlier exposure. Turning to inventors, we find modest effects on green innovation, with stronger responses to late exposure. Our research provides new evidence on how personal experiences shape career trajectories and the evolution of technical progress, advancing our understanding of the factors that shape the supply of skills required to address a changing environment.

**The Effect of Coal Power Retirements on Labor Market Outcomes,** *with Gordon Hanson*

Nearly half of U.S. coal-fired generators operating in 2000 had retired by 2022, representing more than one third of coal capacity. As electricity production shifts elsewhere, do the industries and jobs that depend on it follow? We build a dataset of roughly 1,100 coal generators, linking EIA, EPA, and FERC records on plant characteristics and regulatory exposure. Using LASSO selection models, we identify size, age, and emissions as the key predictors of closure. Retirement is largely a plant-level decision and unfolds gradually over years of declining generation. Replacement of lost capacity is also incomplete. Between 2010 and 2022, plants that retired coal generators lost roughly 80% of their coal capacity and 40% of their total generating capacity. Total capacity in their counties fell nearly 15%, while generation expanded in non-coal counties, more than offsetting the losses. We evaluate the labor market consequences of this local contraction in generating capacity.

**Employment**

Credit Associate, Bridgepoint Capital 2019-2021
Investment Banking Analyst, Barclays Investment Bank 2017-2019

**Affiliations**

Doctoral Student Fellow, Harvard Environmental Economics Program

**Seminars & Conferences**

Harvard/MIT Environmental Economics and Policy Seminar 2026
CERAWeek NextGen Cohort Poster Presentation 2026
Carnegie Mellon University Engineering and Public Policy, Doctoral Student Participatory Workshop on Climate and Energy Decision Making 2026

**Academic Service**

Peer Mentor, Kennedy PhD Student Association (KPSA) 2024-2025
Social Chair, Kennedy PhD Student Association (KPSA) 2022-2023

**Policy Reports**

"The economic geography of salmon: A conceptual framework and preliminary characterization of the spatial distribution of economic values associated with salmon in the Mat-Su Basin."
*Alaska Department of Fish and Game*, 2015 (with David Albert, David Holen, Bronwyn Jones, and Tobias Schwörer)

"Alaskan cod market analysis."
*Bristol Bay Economic Development Corporation*, 2013 (with Matthew Reimer)

**Languages**

English (native)

**Citizenship**

United States

**Software skills**

Stata, MATLAB, ArcGIS, Python, R, LaTeX, Excel